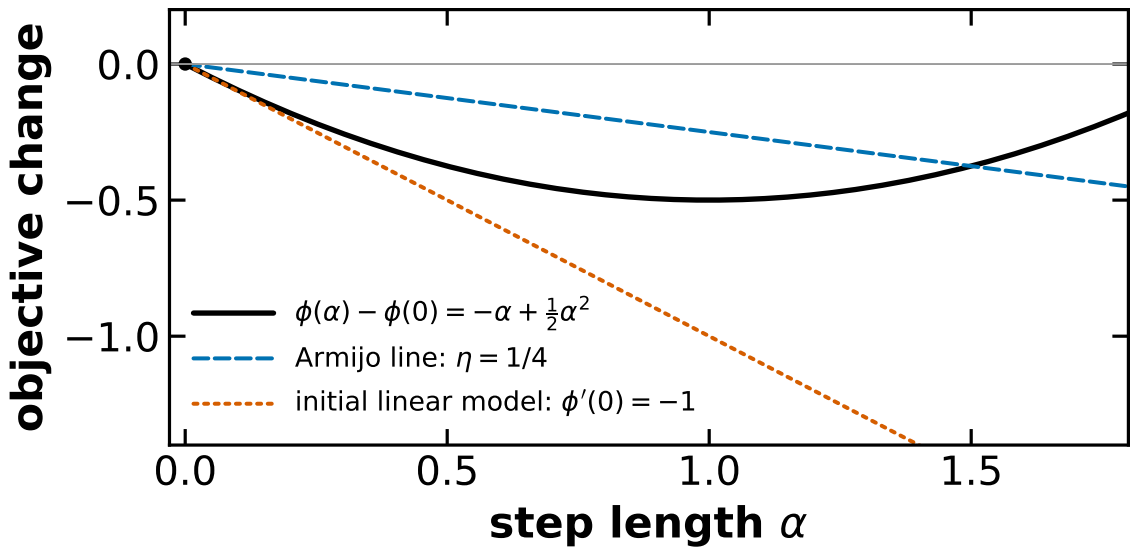
objective change
0.0
−0.5
−1.0
0.0
0.5
1.0
1.5
step length α
$\phi(\alpha) - \phi(0) = -\alpha + \frac{1}{2}\alpha^2$
Armijo line: $\eta = 1/4$
initial linear model: $\phi'(0) = -1$